

# Dhruv Dewan

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## EDUCATION

### University of Maryland

*Bachelor of Science in Computer Science | Bachelor of Science in Mathematics*  
*Gemstone Honors Program*

College Park, MD

*Expected May 2026*

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## EXPERIENCE

### AI Architecture Intern

June 2025 – Present

*Guardian Life Insurance*

*New York, NY*

- Prototyping multi-agent LLM systems with collaborative memory, role-based workflows, and self-evaluation
- Researching model self-reflection in autonomous agent pipelines to identify reliability and alignment concerns
- Built a custom LangChain model for a unified LLM invocation API, enabling seamless switching across foundation models for comparative workflow evaluation

### Machine Learning Engineer Intern

May 2024 – April 2025

*BuildWithin*

*Washington, DC*

- Designed and built a scalable matching search API enabling real-time semantic and keyword-based retrieval across 1000+ active job and candidate profiles on platform
- Optimized model training and reduced inference latency by 90% through batch processing
- Engineered end-to-end MLOps pipeline, including data extraction from PostgreSQL and orchestration of training workflows on cloud infrastructure

### Undergraduate Researcher

Aug 2022 – Present

*Team AID (Artificial Intelligence Diagnosis)*

*College Park, MD*

- Fine-tuned a ResNet34 model for melanoma classification, achieving a 5% absolute accuracy gain on the ISIC test set by incorporating 10% additional diverse skin tone data, improving domain generalization
- Boosted diagnostic accuracy for underrepresented skin tones from an initial 36.2% to 59.4%
- Co-authored a 30-page literature review synthesizing advances in CNN models and GAN-based data augmentation

### Application Development Intern

May 2019 – Sep 2020

*Mokxa Technologies*

*Columbia, MD*

- Engineered a scheduling application for doctor appointments using Joget open-source software, improving appointment management efficiency by 70% through agile development and regular progress presentations
- Completed designer courses, learned Java plug-in development, and created application workflows in Joget
- Co-developed a full-scale web-based application and provided ad-hoc website development support

## PROJECTS

### The Trivia Game | MongoDB, Express.js, Node.js, Git

Oct 2024 - Dec 2024

- Designed and implemented database schemas in MongoDB to efficiently manage and query user game data
- Built and deployed a dynamic leaderboard and ranking system with real-time updates
- Integrated secure user authentication by implementing bcrypt for password hashing ensuring data privacy

### Magnificent 7 ETF Analysis | Python, Pandas, Keras, Pandas-ta, Matplotlib

March 2024 – May 2024

- Developed a comprehensive tutorial on financial stock analysis, feature engineering, and model prediction
- Conducted in-depth technical analysis using lagging indicators to identify trends and potential buy/sell signals
- Built and trained a Long Short-Term Memory neural network model to predict future stock prices

### Vocal Range Enhancer | Git, Python, Pygame, Librosa, Photoshop

Apr 2024

- Developed an innovative application to help enhance users' vocal ranges, providing real-time feedback and control
- Crafted a dynamic user interface using Pygame, allowing users to manipulate the application with their vocals
- Managed project collaboration and version control using Git, ensuring seamless teamwork and code integration

## TECHNICAL SKILLS

**Languages:** Python, Java, C, SQL, JavaScript, HTML/CSS, R, MATLAB, MIPS Assembly

**ML / Data Libraries:** PyTorch, LangChain, HuggingFace, scikit-learn, Pandas, NumPy, Keras, Matplotlib

**MLOps / Deployment:** Docker, FastAPI, TensorBoard, Weights & Biases, Git, Linux/Bash

**Cloud / Data Platforms:** Google Cloud Platform, Snowflake, BigQuery